UNIVERSAL OPENNESS IS TESTED ON AFFINE-SPACE EXTENSIONS AND ON LOCALLY FINITELY PRESENTED BASE CHANGES

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Editorial note. Statement span is the whole lemma environment lines 21725-21737 with the three enumerated conditions; proof span is lines 21739-21785.

Editorial note. This card and S3-morph-char-univ-open-finite are deliberately both counted: the two lemmas are distinct statements in the same section, and the second one's proof reuses the polynomial-square trick rather than restating the first.

Editorial note. The proof contains one xymatrix display; the retained preamble provides the Xy-pic setup.

Editorial note. The proof cites no label and no bibliography entry, so no external_references are needed.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of a universally open morphism, base change of schemes, the principal opens D(g) as a basis, and the affine-space extension A^n x_S X. The two reductions are retained in full.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent*

(1) *f is universally open,*
(2) *for every morphism $S' \to S$ which is locally of finite presentation the base change $X_{S'} \to S'$ is open, and*
(3) *for every n the morphism $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is open.*

Proof. It is clear that (1) implies (2) and (2) implies (3). Let us prove that (3) implies (1). Suppose that the base change $X_T \to T$ is not open for some morphism of schemes $g : T \to S$. Then we can find some affine opens $V \subset S$, $U \subset X$, $W \subset T$ with $f(U) \subset V$ and $g(W) \subset V$ such that $U \times_V W \to W$ is not open. If we can show that this implies $\mathbf{A}^n \times U \to \mathbf{A}^n \times V$ is not open, then $\mathbf{A}^n \times X \to \mathbf{A}^n \times S$ is not open and the proof is complete. This reduces us to the result proved in the next paragraph.

Let $A \to B$ be a ring map such that $A' \to B' = A' \otimes_A B$ does not induce an open map of spectra for some A-algebra A'. As the principal opens give a basis for the topology of $\mathrm{Spec}(B')$ we conclude that the image of $D(g)$ in $\mathrm{Spec}(A')$ is not open for some $g \in B'$. Write $g = \sum_{i=1,\ldots,n} a'_i \otimes b_i$ for some n, $a'_i \in A'$, and $b_i \in B$. Consider the element $h = \sum_{i=1,\ldots,n} x_i b_i$ in $B[x_1, \ldots, x_n]$. Assume that $D(h)$ maps to an open subset under the morphism

$$\mathrm{Spec}(B[x_1, \ldots, x_n]) \longrightarrow \mathrm{Spec}(A[x_1, \ldots, x_n])$$

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in order to get a contradiction. Then $D(h)$ would map surjectively onto a quasi-compact open $U \subset \mathrm{Spec}(A[x_1, \ldots, x_n])$. Let $A[x_1, \ldots, x_n] \to A'$ be the A-algebra homomorphism sending x_i to a'_i. This also induces a B-algebra homomorphism $B[x_1, \ldots, x_n] \to B'$ sending h to g. Since

$$\begin{array}{ccc} \mathrm{Spec}(B[x_1, \ldots, x_n]) & \longleftarrow & \mathrm{Spec}(B') \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A[x_1, \ldots, x_n]) & \longleftarrow & \mathrm{Spec}(A') \end{array}$$

is cartesian the image of $D(g)$ in $\mathrm{Spec}(A')$ is equal to the inverse image of U in $\mathrm{Spec}(A')$ and hence open which is the desired contradiction. $\square$